

1. What is a truth table?

A mathematical table used in logic to determine the possible truth values of a propositional expression. It lists all possible combinations of input values for a set of variables and the corresponding output value of the expression.

2. Which gates are called universal gates and why?

NAND and NOR gates are known as universal gates because every other logic gate can be implemented using only one of these gate types. This property makes them fundamental building blocks for digital circuits.

3. What is the fundamental difference between an HDL (like Verilog) and a software programming language (like C)?

The fundamental difference between an HDL like Verilog and a software programming language like C is that HDL describes the structure and behavior of electronic circuits, while a software language describes a sequence of instructions for a computer's processor. HDL code is written to be synthesized into physical hardware circuits, where operations can occur in parallel. C code is compiled into a series of machine instructions that are executed sequentially.

4. Define "identifier" in the context of Verilog.

-In Verilog, an identifier is a name given to a variable, constant, module, or any other user-defined object. Identifiers must begin with a letter or an underscore, and can be followed by letters, numbers, or underscores. They are also case-sensitive. For example, `my_variable`, `count_2`, and `_temp` are valid identifiers.

5. A 2-input logic circuit has one HIGH and one LOW input, and its output is HIGH. What type of gate is it?

-A 2-input logic circuit with one HIGH and one LOW input that produces a HIGH output could be an OR gate or an XOR gate. An OR gate's output is HIGH if any of its inputs are HIGH. An XOR gate's output is HIGH if the inputs are different.

6. A logic circuit requires all its inputs to be HIGH to make the output HIGH. What type of gate is it?

A logic circuit that requires all its inputs to be HIGH to make the output HIGH is an AND gate.

1. What is meant by "ports" in a Verilog module?

-Ports in a Verilog module define the interface for communication between the module and the outside world. They are like the pins on a physical integrated circuit (IC) that allow signals to enter or leave the module, enabling it to be connected with other modules in a larger design.

2. List the different port modes (e.g., input, output) and their purposes.

-The different port modes and their purposes are:

input: A port that receives signals from the external environment. It's used for data that flows into the module.

output: A port that sends signals out of the module. It's used for results or data produced by the module.

inout: A bidirectional port that can both receive and send signals. It's commonly used for shared data buses where data needs to flow in both directions.

3. Describe the different categories of operators in Verilog (e.g., bitwise, logical).

Logical Operators

Logical operators perform boolean logic on expressions, and their result is always a single bit. They are typically used in conditional statements. They include logical AND, logical OR, and logical NOT.

Bitwise Operators

These operators perform a bit-by-bit operation on two or more operands. The result has the same bit-width as the operands. They include bitwise AND, bitwise OR, bitwise XOR, bitwise NOT, and bitwise XNOR.

Reduction Operators

Reduction operators perform a bitwise operation on all bits of a single vector operand to produce a single-bit result. For example, &A returns the result of ANDing all the bits in vector A. Others include OR, XOR, NAND, and NOR.

Shift Operators

These operators shift the bits of an operand to the left or right. They include left shift and right shift. An example is shifting a binary number to the left to multiply it by a power of two.

Concatenation Operator

The concatenation operator combines two or more smaller operands to create a single, larger vector. It is represented by curly braces.

Conditional Operator

The conditional operator provides a shorthand for an if-else statement. It takes three operands: a condition, a value if the condition is true, and a value if it's false

4. What is the difference between blocking (=) and non-blocking (<=) assignment operators? (Note: We will cover this in detail for sequential logic, but research the basic difference).

The key difference between blocking and non-blocking assignment operators is their execution order:

Blocking Assignments are executed sequentially in the order they appear. A statement with a blocking assignment will not be completed until the previous blocking statement is finished.

This is typically used for modeling combinational logic.

All non-blocking assignments within a procedural block are scheduled to happen at the same time, or in parallel, after all right-hand side expressions are evaluated. This is essential for modeling sequential logic like flip-flops, as it prevents race conditions.

5. What is meant by "simulation" in the context of digital design?

In digital design, simulation is the process of using software to model the behavior of a digital circuit. It involves applying a sequence of inputs to the circuit's description and observing the outputs over time. This virtual testing allows designers to verify that their circuit functions as intended, identify and debug logic errors, and validate the design before it is physically built or synthesized into hardware.

CODE PRESENT IN GITHUB

<https://github.com/heapdats/COE180.1>